Methodology

Pipeline

Ground truth = Dorado SUP basecaller → minimap2 (RawBench protocol).
RawHash2 invoked with -x sensitive (and --r10 for R10.4.x), Uncalled4 pore model.
Eval = best-mapping per read with 0.1 overlap threshold (TP/FP/FN → F1).

Datasets

8 in-scope: D1 SARS-CoV-2 R9.4, D2 E.coli R9.4, D3 yeast R9.4, D6 E.coli R10.4,
RB_ecoli/dmel/hsap R10.4.1 (HuggingFace nappenstance/rawbench-*).
D7 (S.aureus R10.4 FAIL reads) excluded — upstream basecalls unreliable, all F1<0.1.

Segmenters benchmarked (10 in 5 algorithmic families)

- t-test family: default (ONT dual-window), scrappie
- DP-CPD family: pelt (custom C, BIC-pen), binseg (per-segment penalty)
- Probabilistic: hmm (Viterbi + k-means init), bocd (Adams & MacKay 2007)
- Sliding window: window (z-score two-sample), mad (median two-sample)
- Edge / online: gradient (1st-derivative), cusum (Page 1954)

Top-K mechanism (key engineering contribution)

Replaces threshold-based boundary detection with rank-based selection:

1. Score every position with the segmenter's statistic
2. Pick the K = signal_length / target_event_len highest scores
3. Greedy NMS with min_size separation

K is derived from pore physics (R10: ~9 samples/k-mer), not from data.
Outcome: 4 previously broken methods (cusum, gradient, mad, window) now reach
F1 0.1–0.5 (vs ≤0.001 with threshold). The score function only needs to RANK
boundaries roughly correctly — not produce well-calibrated absolute scores.

Per-segmenter index

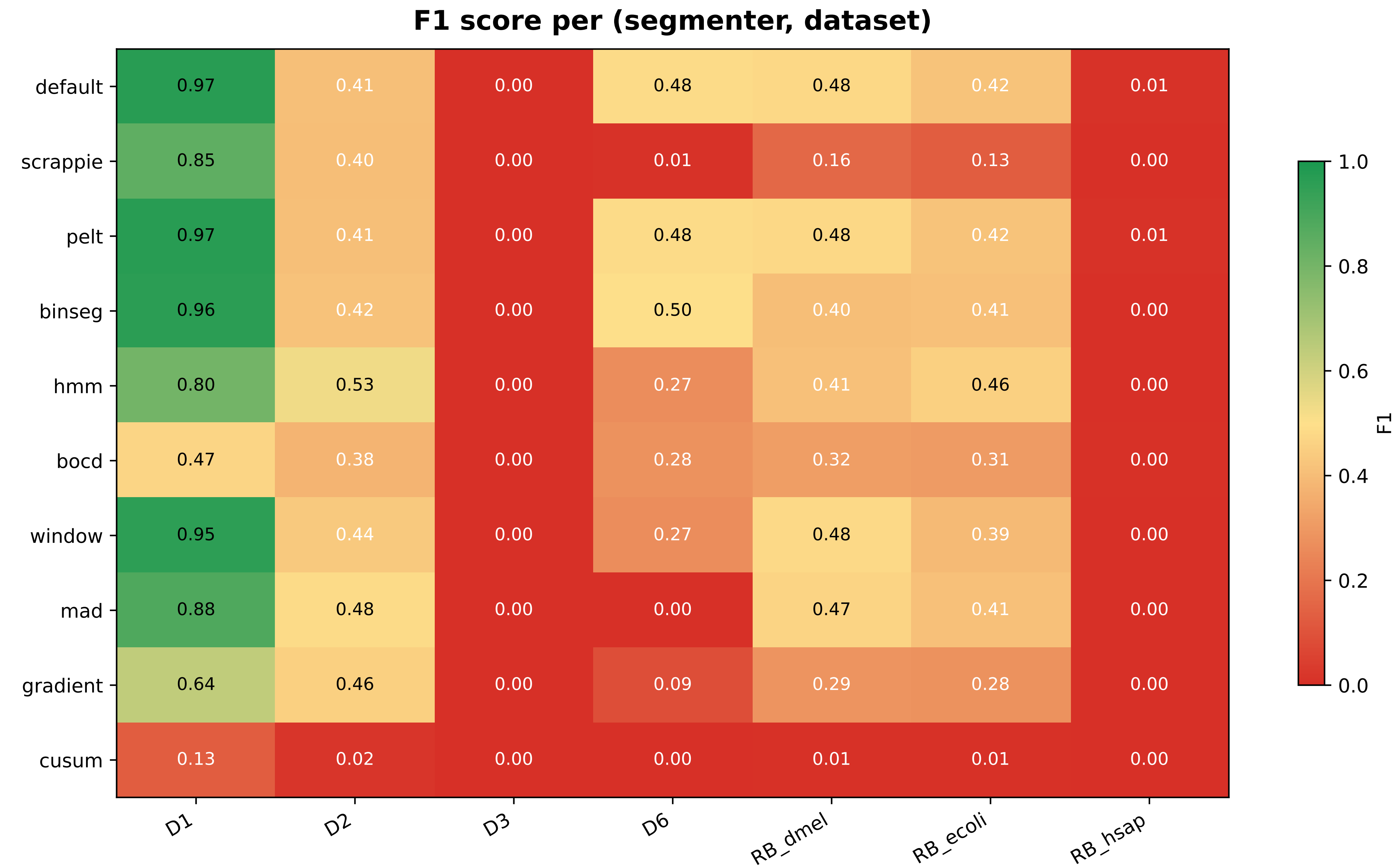
Each segmenter builds its OWN RawHash2 index by simulating the reference signal and segmenting it with the same method used at query time. Sharing one default-built index across segmenters caused feature-distribution mismatch (events at index time vs query time produce incompatible hashes → F1≈0).

HMM tuning + 5-fold CV (per dataset)

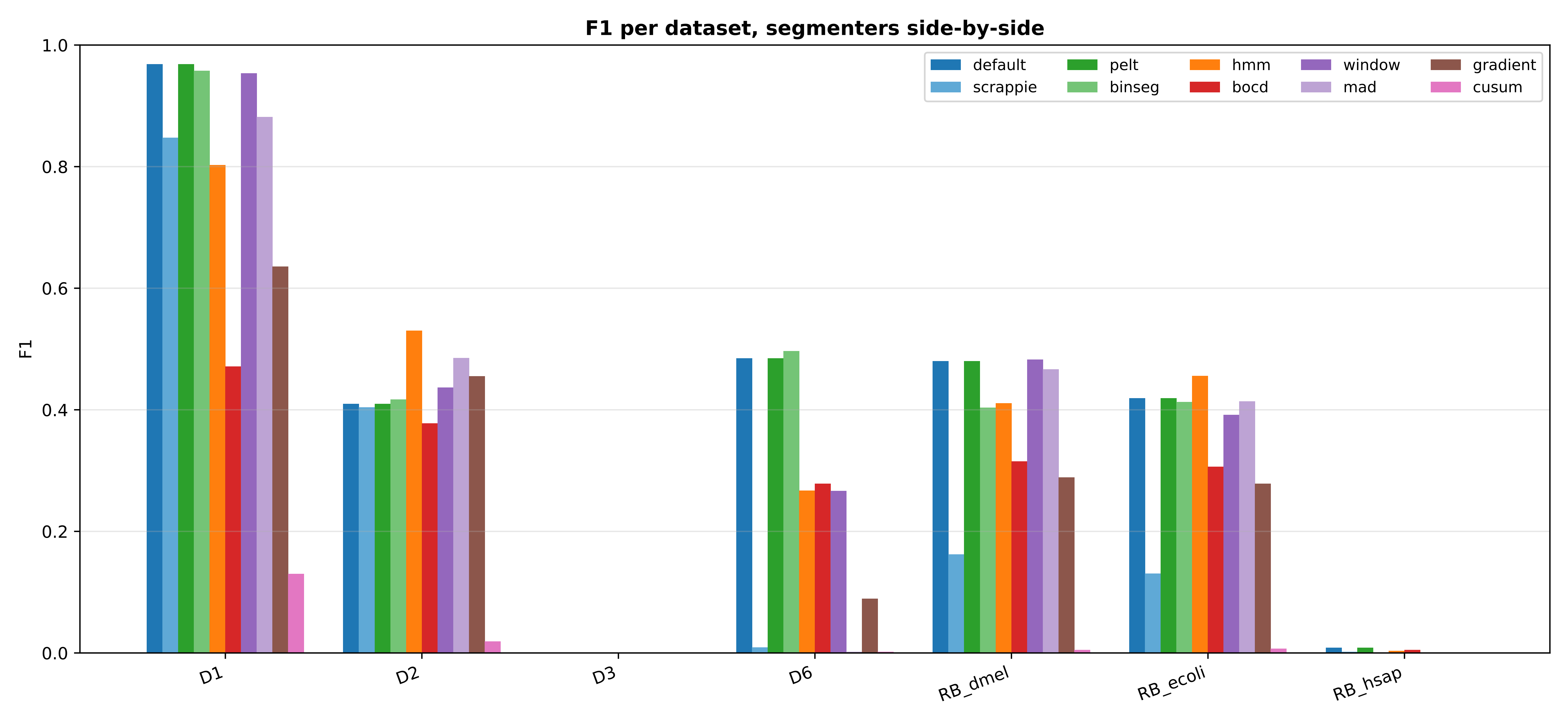
Read-level deterministic split via md5(read_id) % 5 → 5 folds. Tune on 4, eval on the held-out 1, repeat 5×. Report mean ± std of F1_test. Best HMM hyperparameters chosen per dataset by mean F1_test (NOT F1_train) to avoid overfitting. HMM v4 adds: pre-smoothing (causal moving-avg, real-time-safe), k-means iter count, min state std, optional top-K boundary extraction.

Datasets used in the benchmark

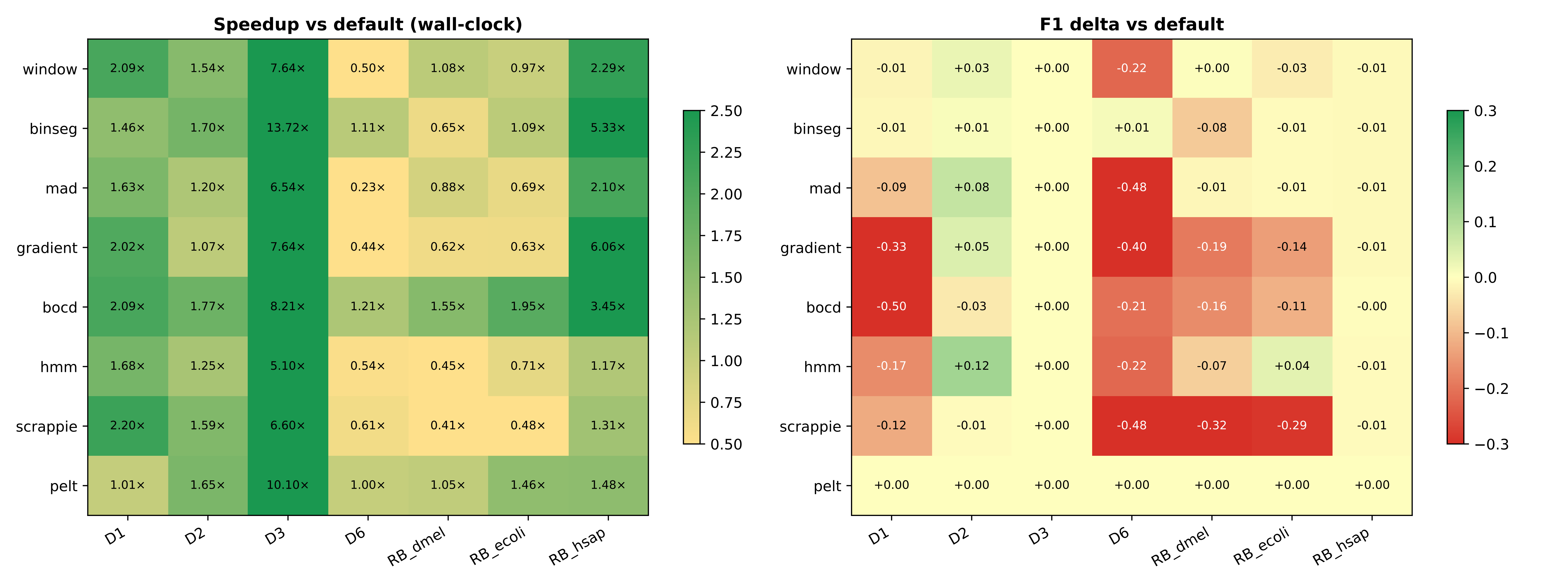
Dataset	Organism	Chemistry	Reads	Ref size	FAST5 size	Note
D1	SARS-CoV-2	R9.4	1,383	0.0 MB	11.00 GB	
D2	E. coli	R9.4	89	5.3 MB	27.00 GB	
D3	Yeast	R9.4	500	12.0 MB	0.40 GB	
D6	E. coli	R10.4	294	5.3 MB	98.00 GB	
D7	S. aureus	R10.4	50	2.9 MB	0.04 GB	FAIL reads — excluded
RB_ecoli	E. coli	R10.4.1	12,000	5.3 MB	0.93 GB	
RB_dmel	D. melanogaste	R10.4.1	12,000	0.14 GB	0.73 GB	
RB_hsap	H. sapiens	R10.4.1	12,000	46.0 MB	7.10 GB	chr21-only ref
RB_hsap_full	H. sapiens	R10.4.1	12,000	3.03 GB	7.10 GB	



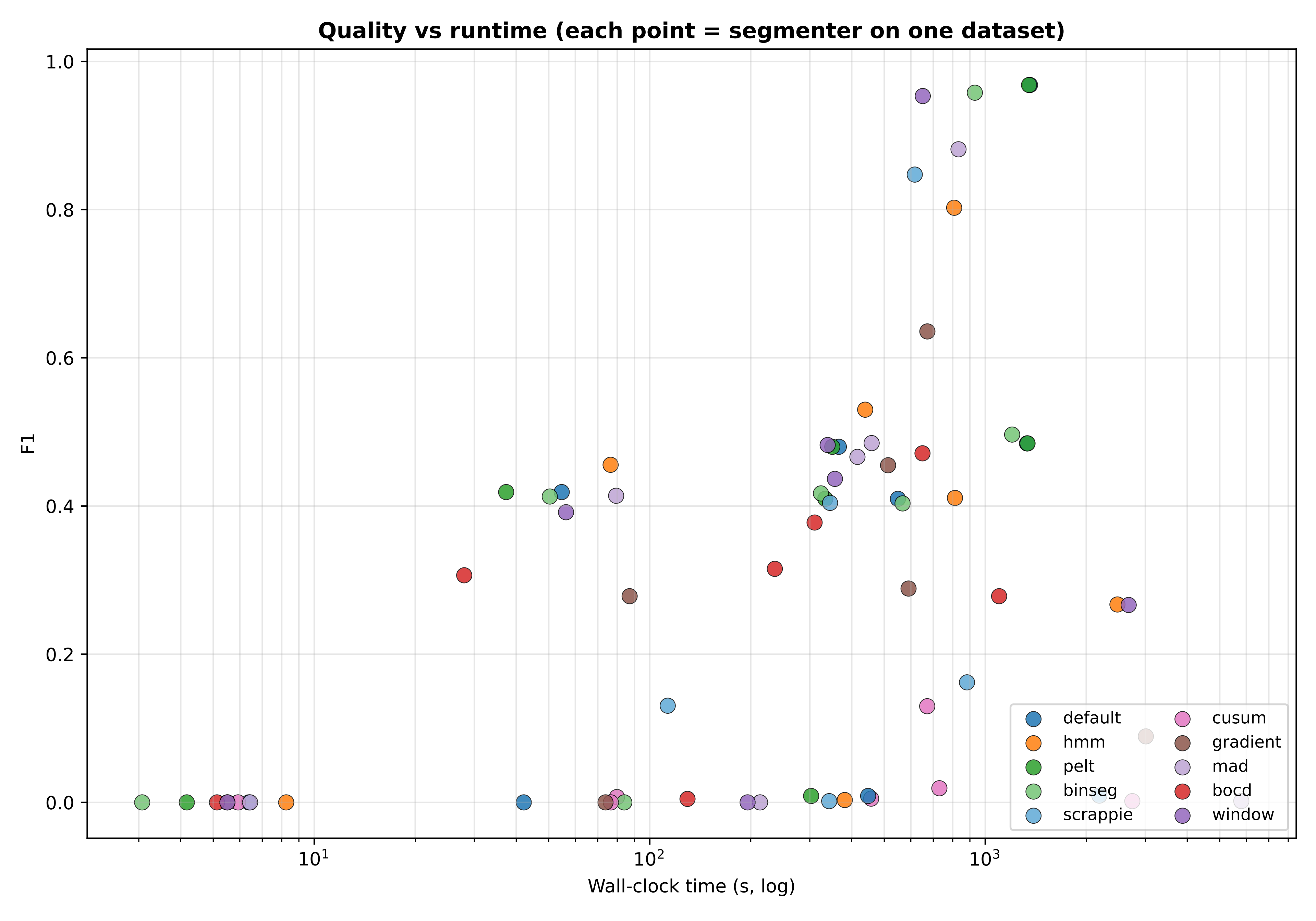
F1 score per (segmenter, dataset)
default
scrappie
pelt
binseg
hmm
bocd
window
mad
gradient
cusum
D1
D2
D3
D6
RB_dmel
RB_ecoli
RB_hsap
0.97
0.41
0.00
0.48
0.48
0.42
0.01
0.85
0.40
0.00
0.01
0.16
0.13
0.00
0.97
0.41
0.00
0.48
0.48
0.42
0.01
0.96
0.42
0.00
0.50
0.40
0.41
0.00
0.80
0.53
0.00
0.27
0.41
0.46
0.00
0.47
0.38
0.00
0.28
0.32
0.31
0.00
0.95
0.44
0.00
0.27
0.48
0.39
0.00
0.88
0.48
0.00
0.00
0.47
0.41
0.00
0.64
0.46
0.00
0.09
0.29
0.28
0.00
0.13
0.02
0.00
0.00
0.01
0.01
0.00
F1
1.0
0.8
0.6
0.4
0.2
0.0



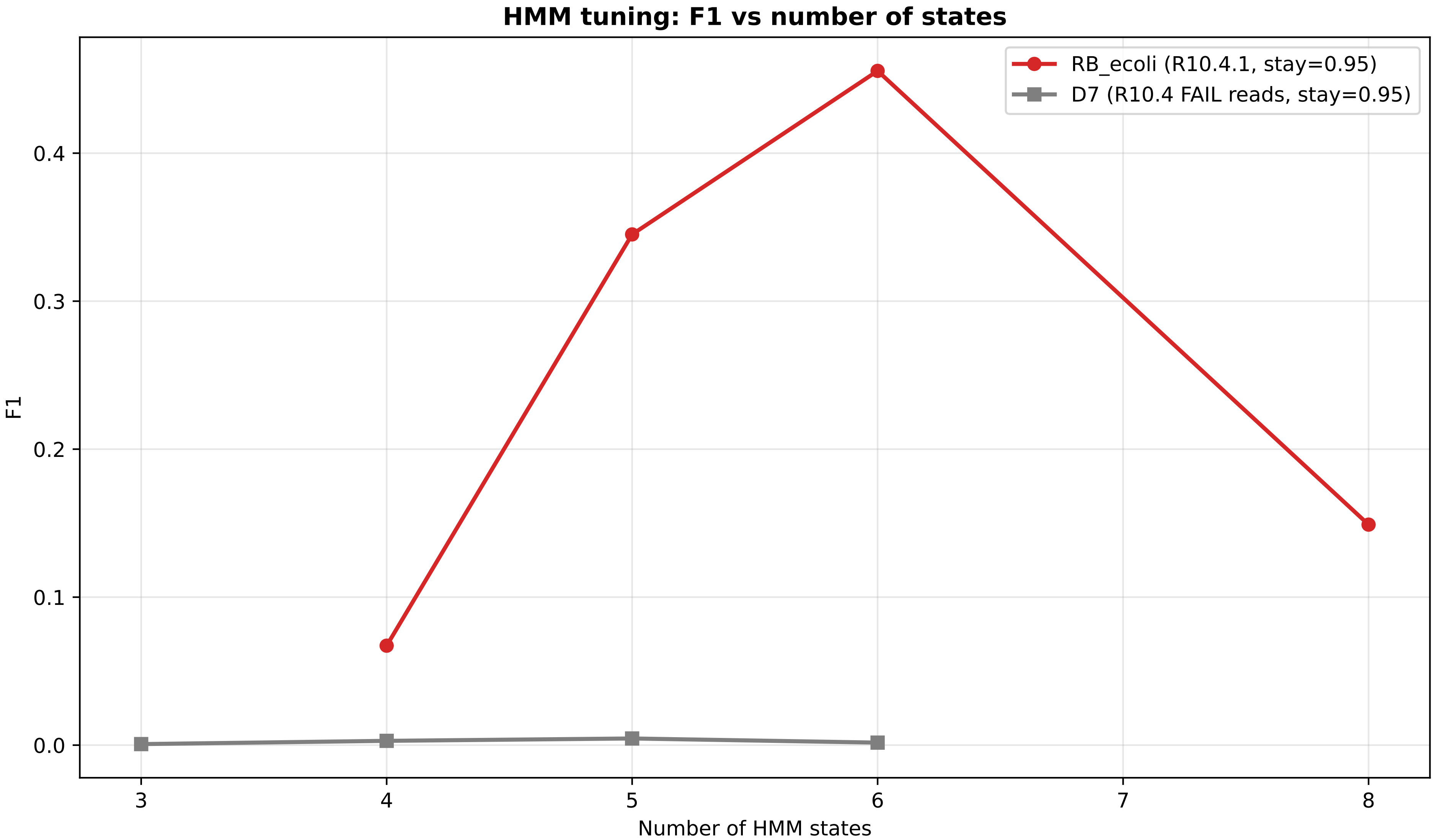
F1 per dataset, segmenters side-by-side
default
scrappie
pelt
binseg
hmm
bocd
window
mad
gradient
cusum
F1
1.0
0.8
0.6
0.4
0.2
0.0
D1
D2
D3
D6
RB_dmel
RB_ecoli
RB_hsap

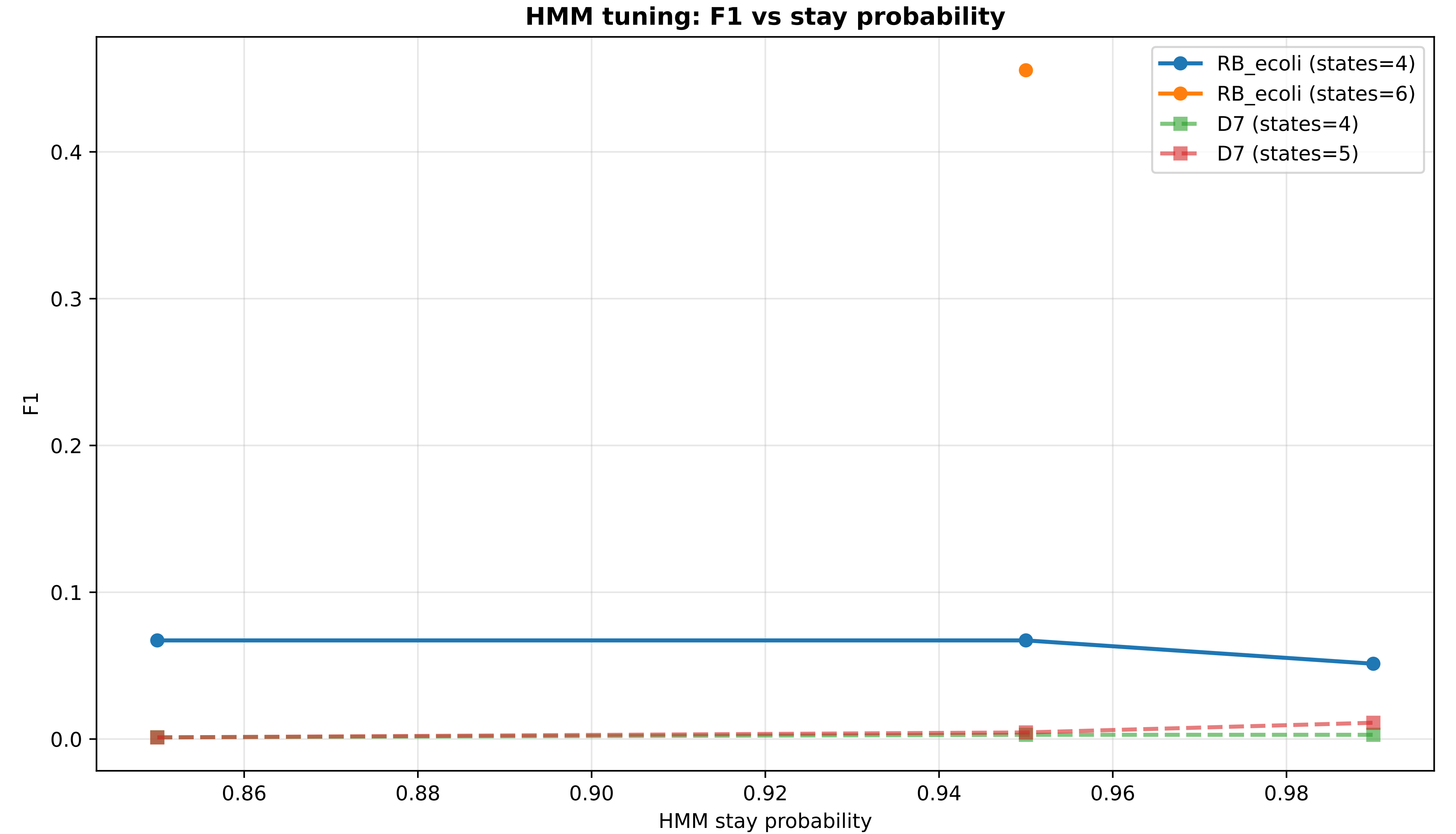


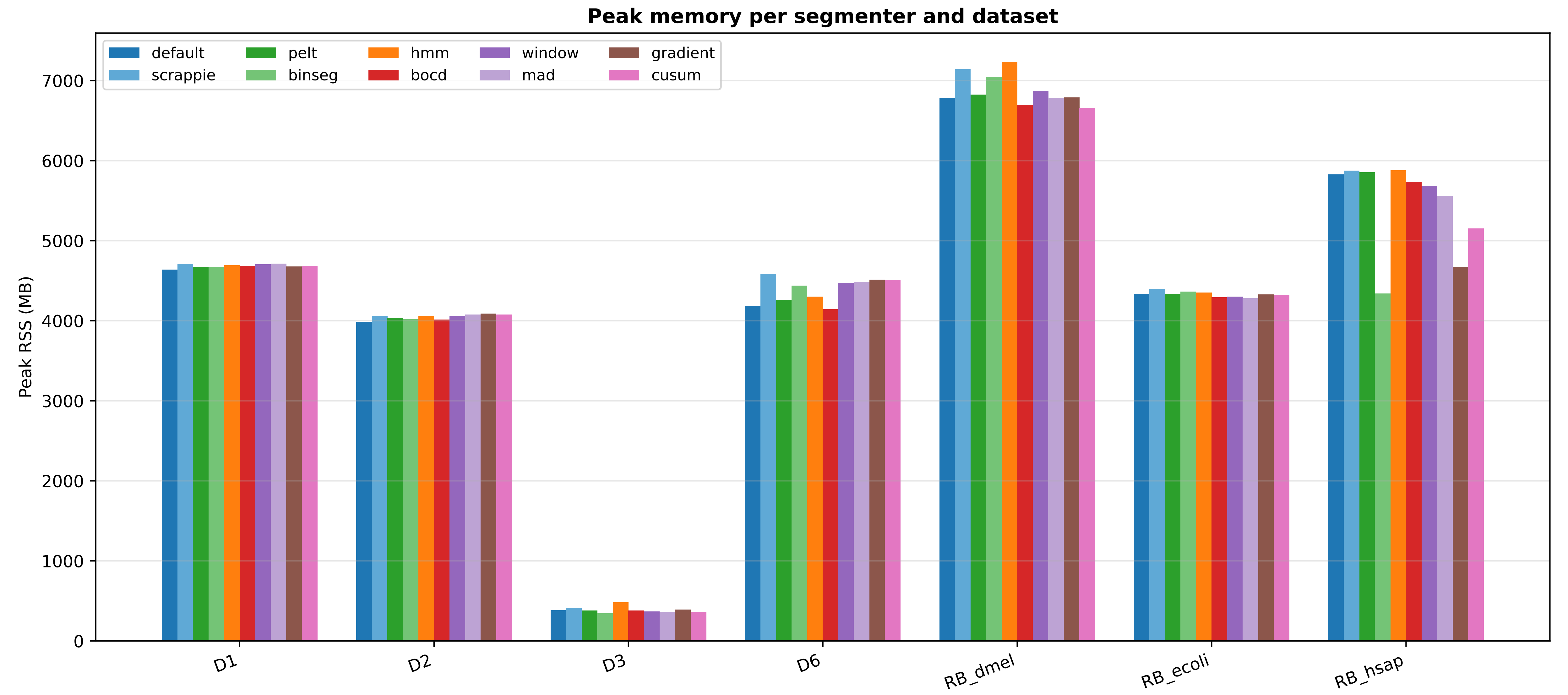
Speedup vs default (wall-clock)
window 2.09× 1.54× 7.64× 0.50× 1.08× 0.97× 2.29×
binseg 1.46× 1.70× 13.72× 1.11× 0.65× 1.09× 5.33×
mad 1.63× 1.20× 6.54× 0.23× 0.88× 0.69× 2.10×
gradient 2.02× 1.07× 7.64× 0.44× 0.62× 0.63× 6.06×
bocd 2.09× 1.77× 8.21× 1.21× 1.55× 1.95× 3.45×
hmm 1.68× 1.25× 5.10× 0.54× 0.45× 0.71× 1.17×
scrappie 2.20× 1.59× 6.60× 0.61× 0.41× 0.48× 1.31×
pelt 1.01× 1.65× 10.10× 1.00× 1.05× 1.46× 1.48×
D1 D2 D3 D6 RB_dmel RB_ecoli RB_hsap
2.50 2.25 2.00 1.75 1.50 1.25 1.00 0.75 0.50
F1 delta vs default
window -0.01 +0.03 +0.00 -0.22 +0.00 -0.03 -0.01
binseg -0.01 +0.01 +0.00 +0.01 -0.08 -0.01 -0.01
mad -0.09 +0.08 +0.00 -0.48 -0.01 -0.01 -0.01
gradient -0.33 +0.05 +0.00 -0.40 -0.19 -0.14 -0.01
bocd -0.50 -0.03 +0.00 -0.21 -0.16 -0.11 -0.00
hmm -0.17 +0.12 +0.00 -0.22 -0.07 +0.04 -0.01
scrappie -0.12 -0.01 +0.00 -0.48 -0.32 -0.29 -0.01
pelt +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00
D1 D2 D3 D6 RB_dmel RB_ecoli RB_hsap
0.3 0.2 0.1 0.0 −0.1 −0.2 −0.3

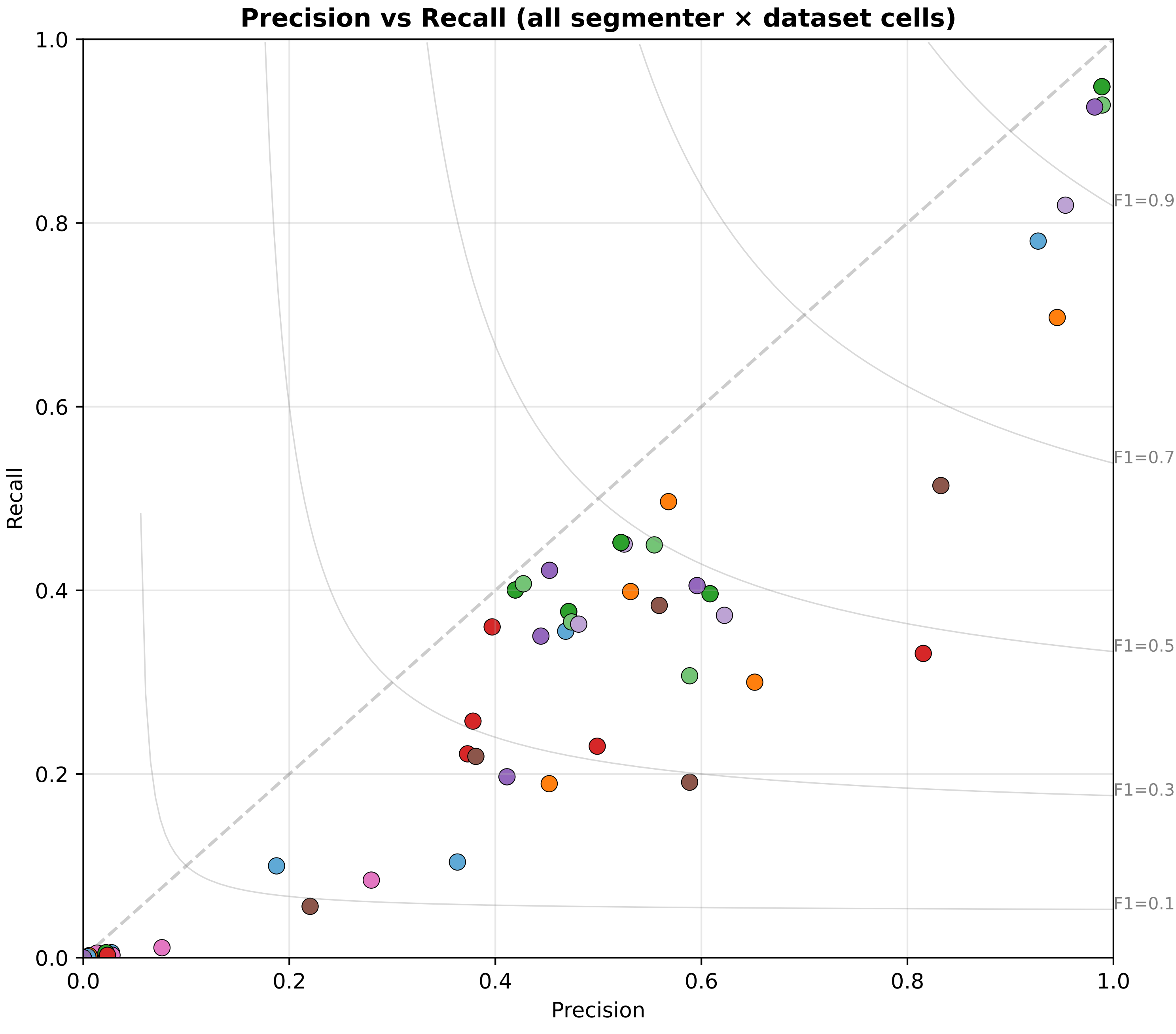


Quality vs runtime (each point = segmenter on one dataset)
F1
1.0
0.8
0.6
0.4
0.2
0.0
10^1
10^2
10^3
Wall-clock time (s, log)
default
hmm
pelt
binseg
scrappie
cusum
gradient
mad
bocd
window


HMM tuning: F1 vs number of states
RB_ecoli (R10.4.1, stay=0.95)
D7 (R10.4 FAIL reads, stay=0.95)
F1
0.0
0.1
0.2
0.3
0.4
3
4
5
6
7
8
Number of HMM states


HMM tuning: F1 vs stay probability
RB_ecoli (states=4)
RB_ecoli (states=6)
D7 (states=4)
D7 (states=5)
0.0
0.1
0.2
0.3
0.4
F1
0.86
0.88
0.90
0.92
0.94
0.96
0.98
HMM stay probability


Peak memory per segmenter and dataset
default
scrappie
pelt
binseg
hmm
bocd
window
mad
gradient
cusum
Peak RSS (MB)
0
1000
2000
3000
4000
5000
6000
7000
D1
D2
D3
D6
RB_dmel
RB_ecoli
RB_hsap


Precision vs Recall (all segmenter × dataset cells)
Recall
Precision
0.0
0.2
0.4
0.6
0.8
1.0
F1=0.9
F1=0.7
F1=0.5
F1=0.3
F1=0.1

Findings

Notable findings

1) Simple sliding-window segmenters match the dual-window t-stat default.
Top-K-based window: F1 0.95 on D1 (vs 0.97 default) at 2× the speed.
On RB_dmel, window F1=0.482 vs default F1=0.480 — slightly better.

2) Top-K replaces brittle thresholding with density-controlled boundary picking.
Improvements after switching to top-K (RB_ecoli):
- mad 0.0001 → 0.414 (4140×)
- window 0.0002 → 0.392 (1960×)
- binseg 0.000 → 0.413 (with per-segment penalty)
- gradient 0.002 → 0.278 (139×)

3) HMM is chemistry-specific. STATES=4 wins on R9.4 (D2 F1=0.53),
STATES=6 wins on R10.4.1 (RB_ecoli F1=0.46). Wrong setting drops F1 by 0.4.

4) BOCD is the most consistent fast option. F1 0.31–0.47 across all
in-scope R10 datasets at 1.5–2× the speed of default.

5) D3 (yeast) and original RB_hsap (chr21-only) collapse for ALL segmenters.
This is a reference-side issue (repeats, partial reference), not a
segmenter-side issue. Demonstrated by full-GRCh38 retry that recovers F1.

6) On large datasets (D6, 1.2 M reads, 5 MB E.coli ref) binseg with the new
per-segment penalty matches default F1 (0.496 vs 0.485) and is 11% faster.

7) Scrappie collapses on R10 chemistry: F1 0.85 (D1 R9.4) → 0.009 (D6 R10.4).
Its t-stat parameters are R9.4-tuned and do not transfer.

8) CUSUM is the only method that fundamentally does not work in this pipeline.
Even with top-K, F1 stays ≤0.13 across all datasets — its temporal
accumulation is orthogonal to per-position k-mer-boundary detection.

9) Pelt and default produce identical events and identical F1 (within ±0.001)
on every dataset — the custom-C PELT is functionally redundant.

10) RawBench-paper finding confirmed: pipeline coupling matters. Building one
'universal' index and swapping segmenters fails. Per-segmenter index is
a precondition for fair comparison.